Supplementary material

Machine learning model for predicting the risk of childhood stunting in Peru

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Supplementary Table S1: Predictor variables of the model grouped by domain (45 variables; ENDES code in parentheses).

Domain (*n*)	Variables
Child (7)	Sex (HC27), Age in months (HC1), Birth weight (M19), Size at birth reported by the mother (M18), Altitude-adjusted hemoglobin (HC56), Interval since the previous birth (HC63), Usual resident (HV102)
Mother, pregnancy and breastfeeding (11)	Mother's education (HC62), Prenatal visits (M14), Month of first prenatal visit (M13), Days of iron during pregnancy (M46), Antiparasitic drugs during pregnancy (M60), Place of delivery (M15), Cesarean delivery (M17), Postpartum vitamin A (M54), Hours until breastfeeding initiation (M34), Formula in the first days (M55G), Duration of breastfeeding (M4)
Household and territory (8)	Wealth index (HV271), Area of residence (HV025), Altitude (HV040), Household members (HV012), Children under five in the household (HV014), Age of household head (HV220), Sex of household head (HV219), Traditional transport (SH77F)
Water and sanitation (10)	Water all day (SH42), Time to water source (HV204), Water treatment (HV237), Boiled water (HV237A), Bottled water (HV237G), Water stored in container (SH48), Container type (SH49), Container with lid (SH50), Shared toilet facility (HV225), Iodine in salt (HV234)
Housing (9)	Rooms (SH71), Bedrooms (HV216), Lighting source (SH70), Kitchen smoke outlet (HV240), Kitchen in separate room (HV242), Other cooking fuel (SH63), Glass windows (SH76B), Wooden windows (SH76C), Windows with screens (SH76D)

Supplementary Table S2: Configuration of the algorithms compared. Hyperparameters were set in advance, without a systematic search; those not listed keep the default values of each library. `random_state = 42` in all models.

Model	Configuration
Logistic regression	L2 regularization; balanced class weights; maximum of 500 iterations
Decision tree (CART)	Maximum depth of 5; balanced class weights
XGBoost	300 trees; `scale_pos_weight` equal to the ratio of negative to positive class; `min_child_weight` $= 10^{-7}$
CatBoost	300 iterations; balanced class weights; native categorical variables
Neural network (MLP)	Three hidden layers of 200 neurons; up to 50 iterations with early stopping; no weighting
LightGBM	300 trees; balanced class weights; native categorical variables

Software: Python 3.12, LightGBM 4.6.0, XGBoost 3.2.0, CatBoost 1.2.10, scikit-learn 1.8.0 and SHAP 0.52.0.

Supplementary Table S3: Constructs and items of the relevance assessment instrument. Items were rated on a five-point Likert scale (1 = strongly disagree; 5 = strongly agree). The instrument was administered in Spanish; the items are translated here.

Construct	Source	Items
Job relevance	TAM2 [1]	**R1.** In my field of work, using a system like this would be important. **R2.** In my field of work, using a system like this would be relevant for deciding on nutritional interventions.
Output quality	TAM2 [1]	**C1.** The risk factors it identifies by department are consistent with my professional experience. **C2.** I am confident that the risk levels assigned by the system are accurate.
Result demonstrability	TAM2 [1]	**D1.** I understand why the system assigns a risk level to a specific child. **D2.** I could explain the results provided by the system to a colleague.
Perceived usefulness	TAM [2]	**U1.** The system would allow me to prioritize which children to attend first when resources are limited. **U2.** Using it would save time compared with current methods of case identification. **U3.** Overall, it would improve the effectiveness of targeting work.
Perceived ease of use	TAM [2]	**F1.** Entering a child's data into the system is easy. **F2.** The risk level and the departmental map are easy to read. **F3.** Learning to use the system would not require extensive training.
Perceptions of external control	TAM3 [3]	**K1.** Except for hemoglobin, the data requested by the system are available in the records of my field of work. **K2.** Hemoglobin data are usually available for the children I attend. **K3.** My institution has the means (equipment, connectivity, staff) to use a tool like this.
Subjective norm	TAM2 [1]	**S1.** My colleagues would consider the use of a tool like this appropriate. **S2.** The health authorities in my field would support its adoption.
Intention to use	TAM [2]	**I1.** I would use this system if it were available in my field of work. **I2.** I would recommend its use to colleagues in nutrition or public health.
Overall rating	—	From 1 to 10, what potential do you assign to this tool for public health work?
Open-ended questions	—	(1) What additional variables or information should the system include? (2) What limitations do you identify from your professional experience? (3) In what specific scenario of your practice would you apply this tool? (4) Recommendations for a next version.

The form also recorded the evaluator's profession, academic degree, years of clinical experience, institution and region; these data were used only to describe the panel.

Supplementary Text S1. Check of the variable selection

To rule out that the feature selection had removed useful information, the effect of adding back the excluded variables was checked. Of the 78 columns available in the preprocessed dataset, 33 were not among the 45 selected. These were ranked by their relationship with stunting: for categorical attributes, the range between the highest and lowest prevalence among categories with at least 500 observations was used, whereas for continuous variables the statistic $2\,|\mathrm{AUC} - 0.5|$ was used.

Four experimental configurations were compared: the 45-variable model and three that added the top-ranked variables (water, sanitation and housing; territory; and both), keeping the partition, the weights, the deterministic seed and the classifier hyperparameters constant.

The ranking identified ten attributes with a marked relationship with stunting, among them the predominant material of the walls (HV214), floor (HV213) and roof (HV215); the source of drinking water (HV201); the

department of residence (HV024); the residual chlorine concentration in water (SH227); the cooking fuel (HV226); and the type of toilet facility (HV205). These attributes make up the block of water, sanitation, hygiene and material housing conditions, recognized as an underlying determinant of linear growth faltering [4,5] and found among the main predictors in studies based on DHS data [6]. None of them constitutes information leakage, as they describe structural household conditions observable before a home visit.

Adding the ten attributes did not improve any of the three configurations tested: the AUC-ROC (0.8295), sensitivity (76.44%) and precision (36.36%) were identical to those of the 45-variable model, because the model did not use the added variables. These attributes do contain information: a classifier trained only with the water, sanitation and housing block achieved an AUC-ROC of 0.6895. Nor do they duplicate the wealth index, which explained between $\eta^2 = 0.038$ and $\eta^2 = 0.098$ of the variance of each one. The 45-variable subset already represents water, sanitation and housing conditions through nineteen correlated attributes (Supplementary Table S1), so the exclusion of the candidate variables is a methodologically sound decision.

References

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